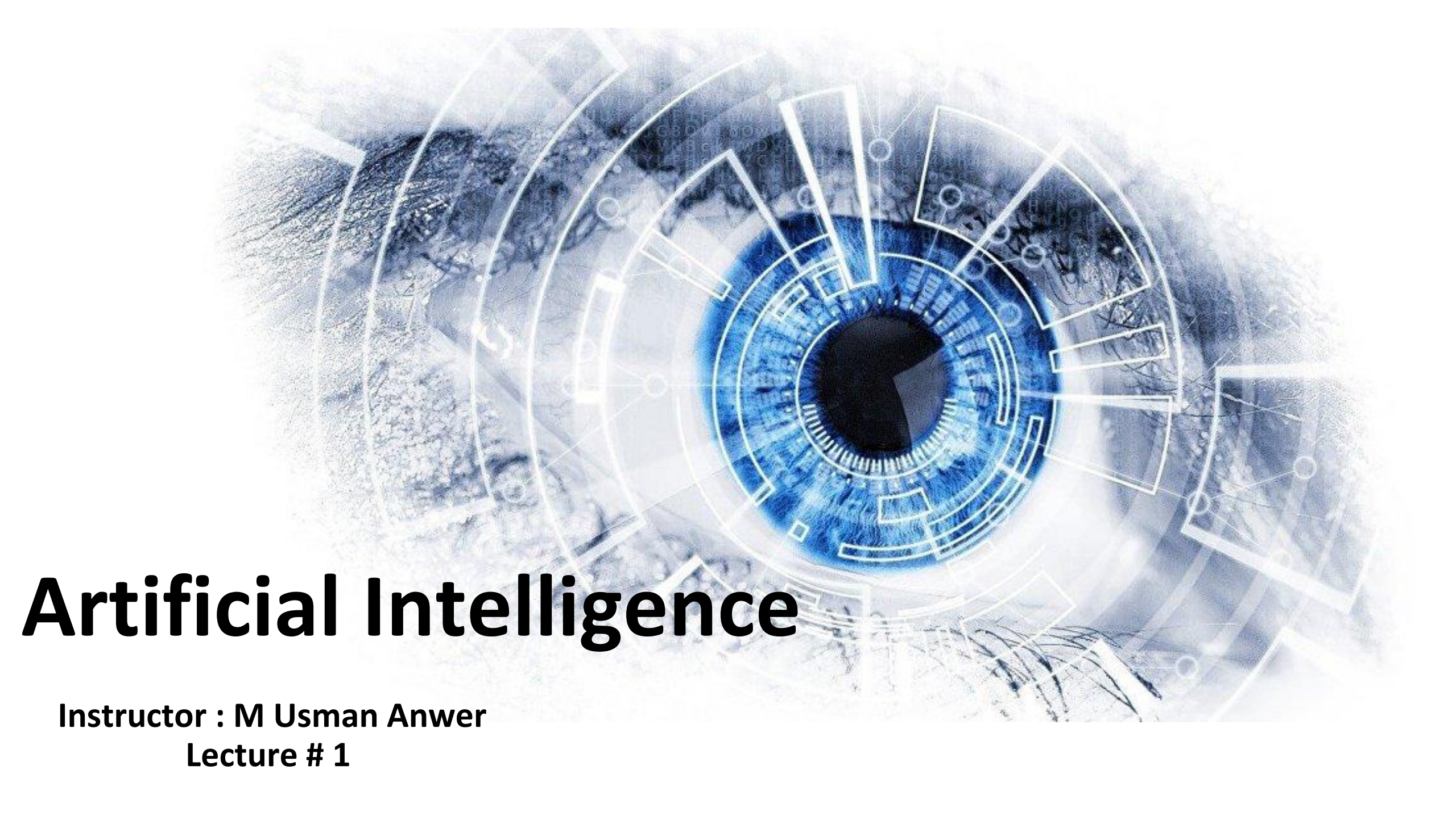




Artificial Intelligence

Instructor : M Usman Anwer
Lecture # 1

Instructor Introduction

- Education:
MSDS, FAST-NUCES, (2021-2024)
- Experience: **10 years**
Industry: Syntax, Upwork (4 years).
Academia: FAST, Bahria, PNY Trainings, NAVTTC (8 years).
Skills: Big Data, Cloud(AWS), MLOps, Mobile Apps
- Interests:
Bioinformatics, Deep Learning, LLMs, MLOps, Cloud Computing

About Myself

Certifications

Cloud:

- AWS Certified Trainer
- AWS Solution Architect (Associate)

MLOps

- MLOps DUKE University
- MLOps Bootcamp Krish Naik

Big Data:

- Big Data Specialization (UC San Diego)
- Cloudera Hadoop(Sundog+Databricks)

Python & Data Science:

- DAT210x: Programming with Python for data science
- Introduction to Python for Data Science
- Introduction to Data Science in Python
- Applied Machine Learning in Python
- Data Science Essentials
- Deep Learning AI

Student's Introduction



Name?

Expectation:

→ What do you expect from this course?

Outline

- Course Overview: Goals & Objectives
- Class Protocols
- Outline of topics to be discussed in the course
- Introduction to AI
- History
- Applications
- Subareas of AI
- Reading Suggestions

What is AI?

- *Artificial intelligence (AI) may be defined as the branch of computer science that is concerned with the automation of intelligent behavior.*
- AI is the **fusion of many**, such as Computer Science, Electrical Engineering, Mathematics, Statistics, Psychology, Linguistics, and Philosophy.
- IBM Research defines Artificial Intelligence (AI) as **Augmented Intelligence**, helping experts scale their capabilities as machines do the time-consuming work.

History of AI

**Sooooooooooooo
BooRRreeeed!**



- 1943 McCulloch & Pitts: Boolean circuit model of brain
- 1950 Turing's "Computing Machinery and Intelligence"
- 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- 1950s Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
- 1965 Robinson's complete algorithm for logical reasoning
- 1966—73 AI discovers computational complexity
Neural network research almost disappears
- 1969—79 Early development of knowledge-based systems
- 1980-- AI becomes an industry
- 1986-- Neural networks return to popularity
- 1987-- AI becomes a science
- 1995-- The emergence of intelligent agents

ALAN TURING
1912 - 1954

Founder of computer science
and cryptographer, whose work
was key to breaking the
wartime Enigma codes,
lived and died here.

ALAN TURING

(1912 - 1954)

“



I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted.

—Mind, vol. 59, #236 (1950)

Alan Turing on Artificial Intelligence

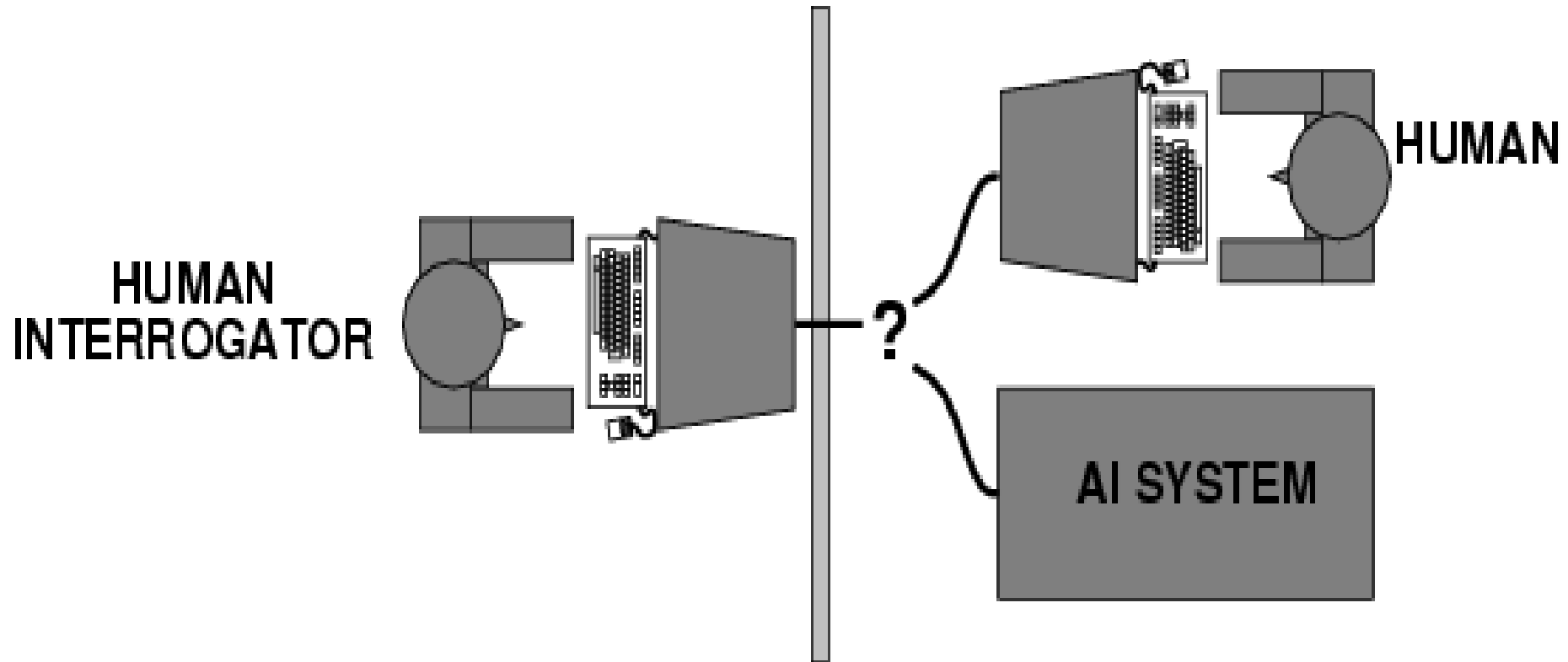
No, I'm not interested in developing a powerful brain. All I'm after is just a mediocre brain, something like the President of the American Telephone and Telegraph Company



A computer would deserve to be called intelligent if it could deceive a human into believing that it was human

Acting humanly: Turing Test

- <http://www.jabberwacky.com/>
- <http://turingtrade.org/>



Can you think of a theoretical system that could beat the Turing test yet you wouldn't find it very intelligent?

Different Types of Artificial Intelligence

Systems that think like humans	Systems that think rationally
Systems that act like humans	Systems that act rationally

note: humans are often not rational!

- Modern AI focuses on the “Act Rationally Approach”
 - modern methods are also inspired by cognitive & neuroscience (how people think).
- This course will also focus on this type.

AI Applications

- **Vehicles:** motion planning for autonomous cars & drones, collision detection
- **Banks:** automatic cheque readers, signature verification systems, automated fake currency detector, automated loan application classification
- **Telephone Companies:** automatic voice recognition for inquiries
- **Credit Card Companies:** automated fraud detection
- **Netflix:** movie recommender
- **Google:** Search Technology, page rank
- **Amazon:** product recommenders, Consumer Marketing, automated help-desk
- **Biometric Identification:** face, eye's iris scan, fingerprints, voice pattern
- **Language Translation:** speech recognition, translation for multiple nationalities,
- **Games:** chess, pokemon, Minecraft, Alphago Zero, Half life, The Last of US,

AI-Applications



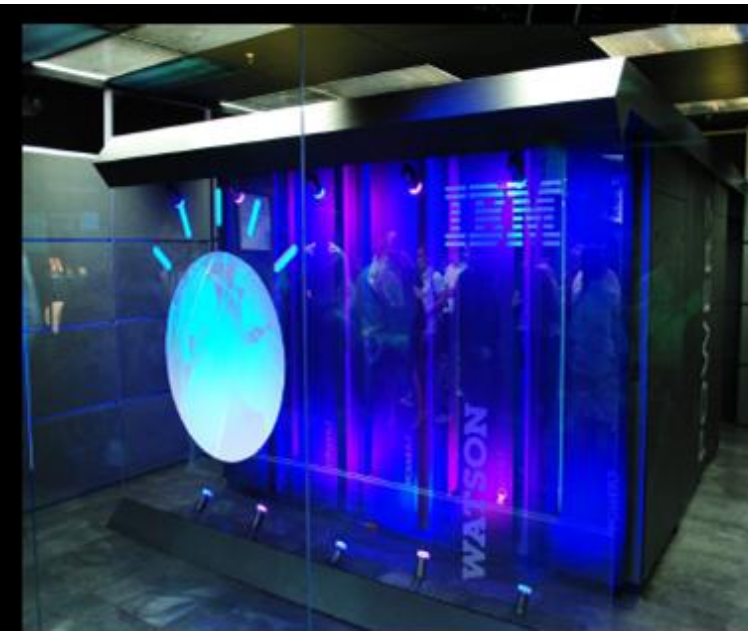
IBM THEN: Project Deep Blue



IBM WATSON



IBM NOW: WATSON



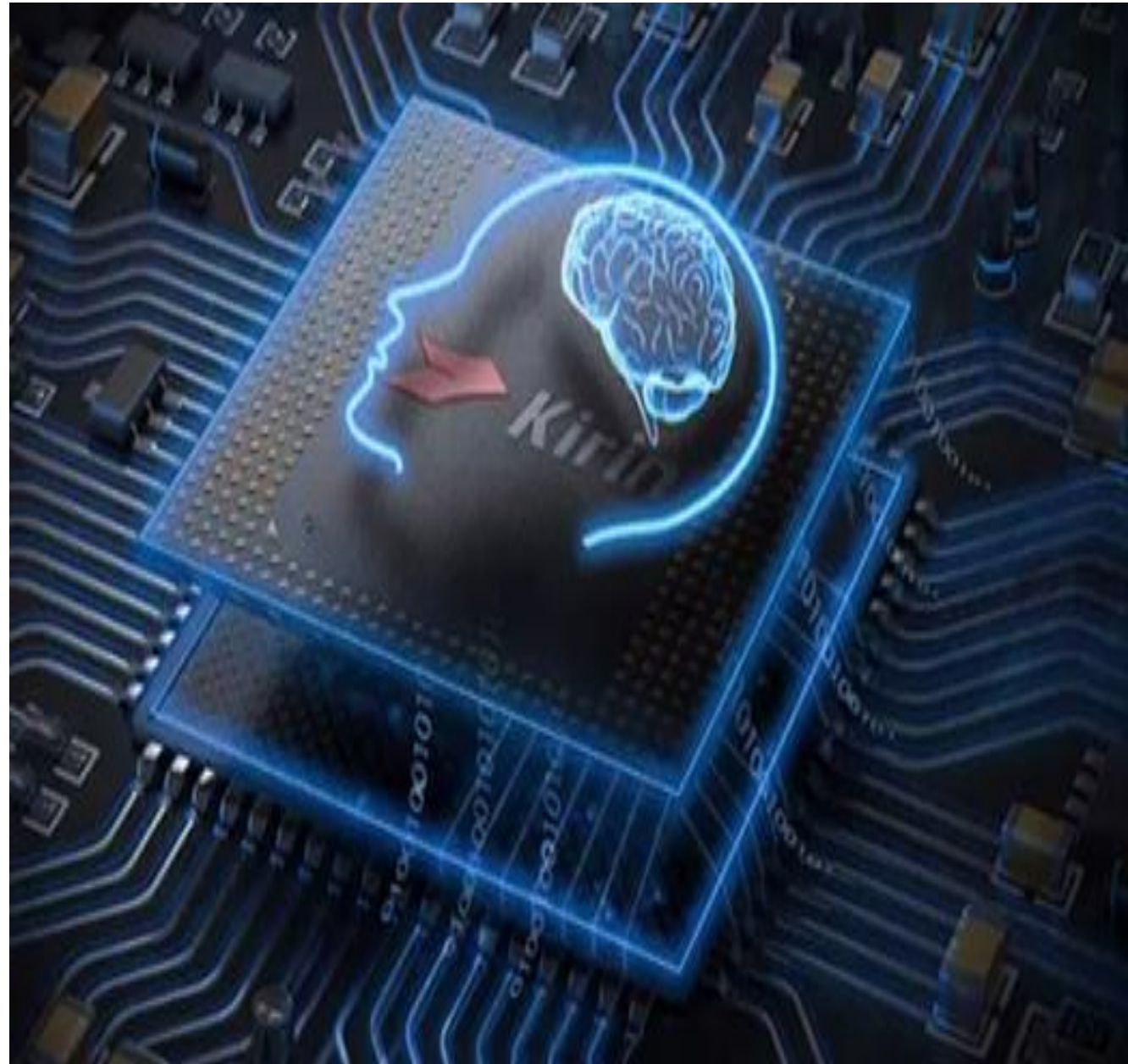
AI-Applications

- Industrial AI

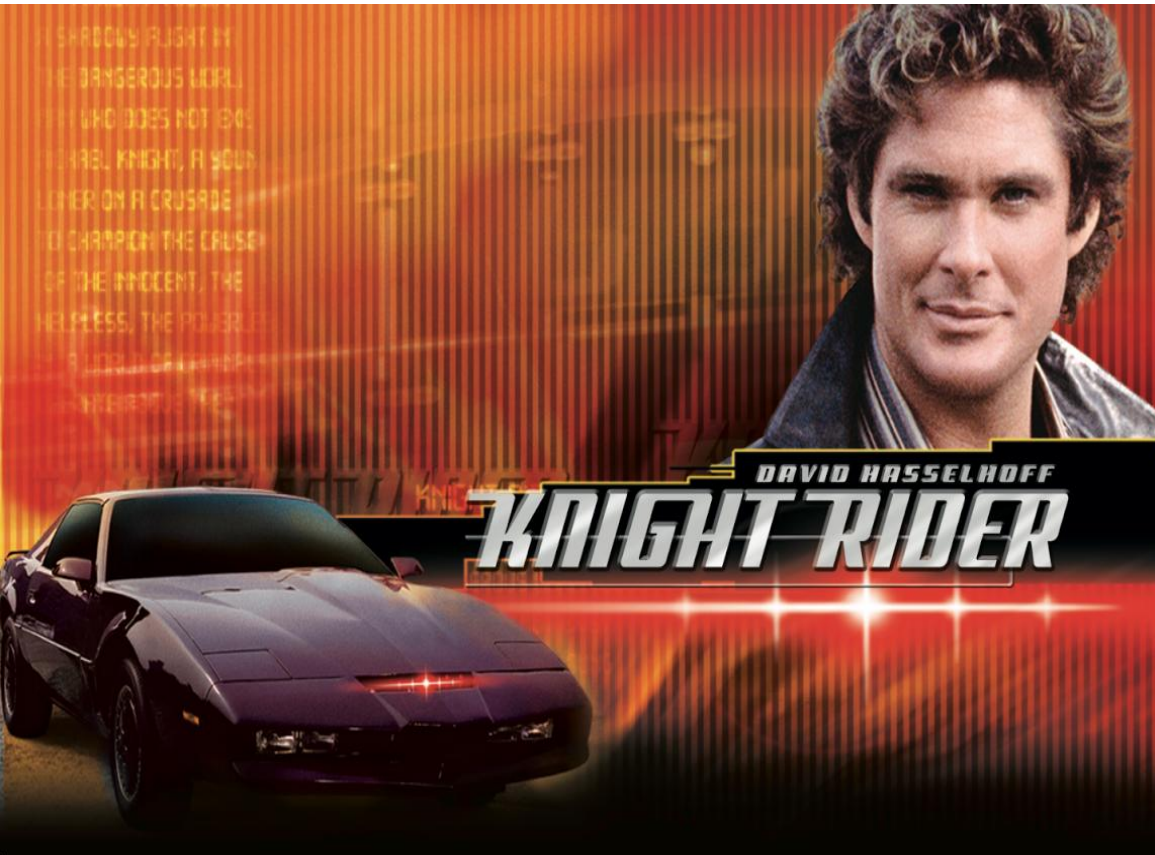


AI in Smart Phones

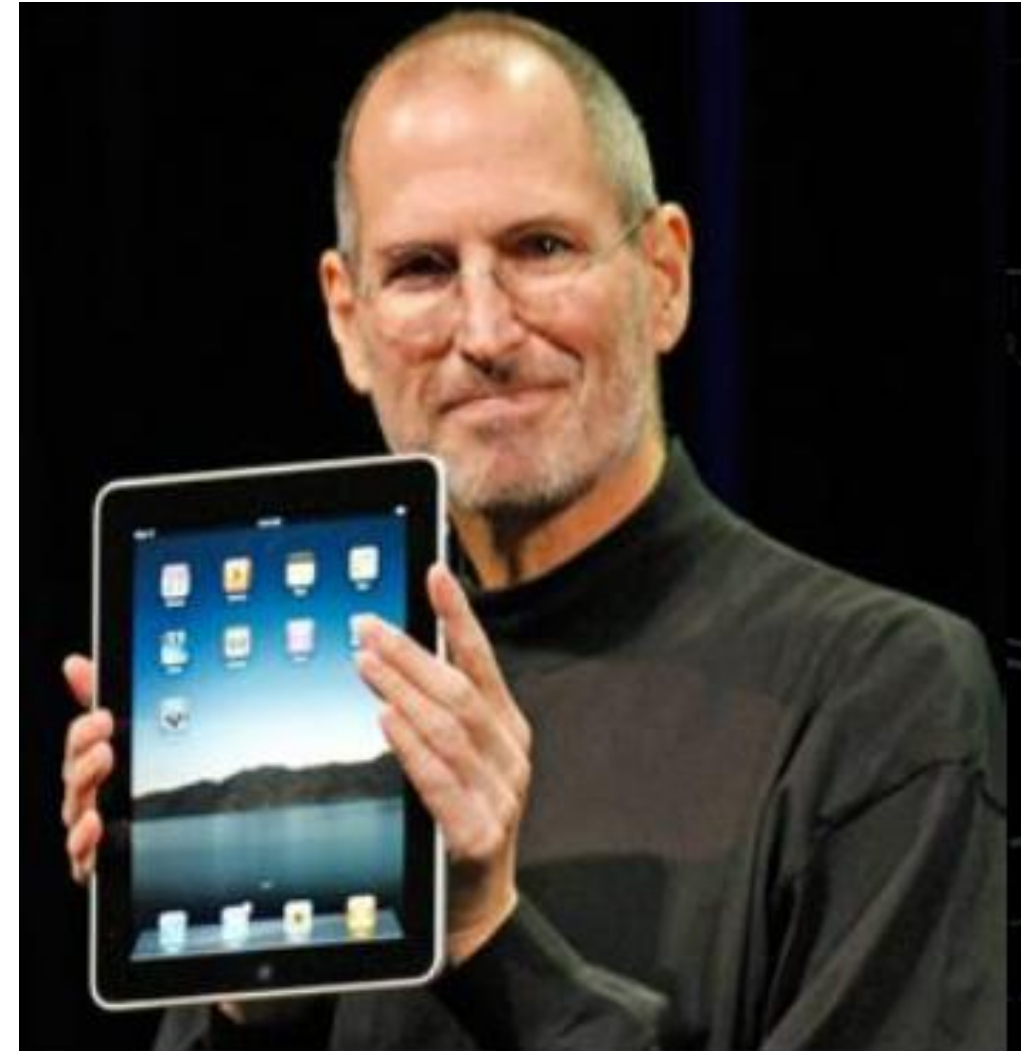
- **iPhoneX** has Neural Engine as part of A11 Bionic chip
- **Huawei Kirin 970** chip has Neural Processing Unit or **NPU** on it
- **Google Pixel2** has a secret AI-powered imaging chip that just got activated. So what exactly are these next-gen chips designed to do?



Fictions Turning into Reality



Fictions Turning into Reality



Fictions Turning into Reality



Fictions Turning into Reality

HAL

- *HAL 2001: A Space Odyssey*
 - classic science fiction movie from 1969
 - An intelligent computer called **HAL** is the “brains” of a spaceship
 - in the movie, HAL can
 - speak easily with the crew
 - see and understand the emotions of the crew
 - navigate the ship automatically
 - diagnose on-board problems
 - make life-and-death decisions
 - display emotions
- Boeing X37-B shows some of its capabilities.



A glimpse into Man-Machine Society



What Next? Fact or Fiction?



What Next?

Year	Events
2015	House hold robots
2018	10 TB = single human brain memory
2020	Nanobots, Humanbody version 2.0
2023	10^{16} calculations per second—roughly the equivalent of one human brain
2029	Capacity to simulate 10 billion neurons
2030	Mind-uploading, Human-body version 3.0, telepathic style communication,
2040	Nonbiological intelligence will be billions of times more capable than biological intelligence Use of Foglets Matrix style life: Virtual world
2045	Artificial intelligences surpass human beings as the smartest and most capable life forms on the Earth

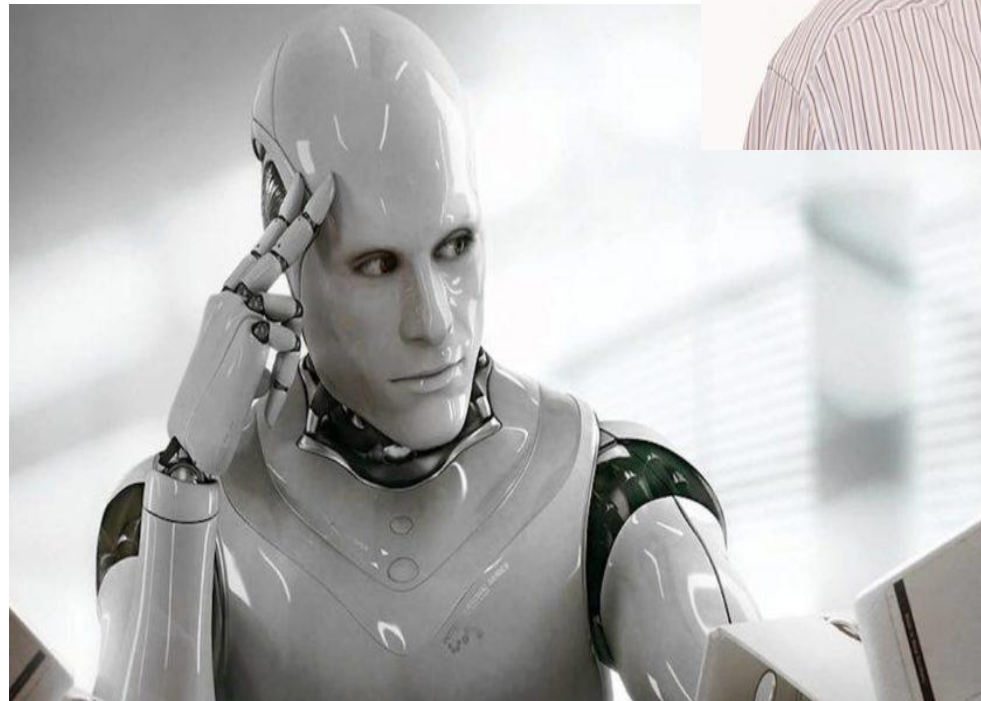
JOBS....

HUMAN NOT APPLY

Is it true?

2045: The Singularity

- Ray Kurzweil
- Humans = Biological + non-biological Intelligence
- AI Will Not Displace Humans, It's Going to Enhance Us



Russian Project Avatar: 2045

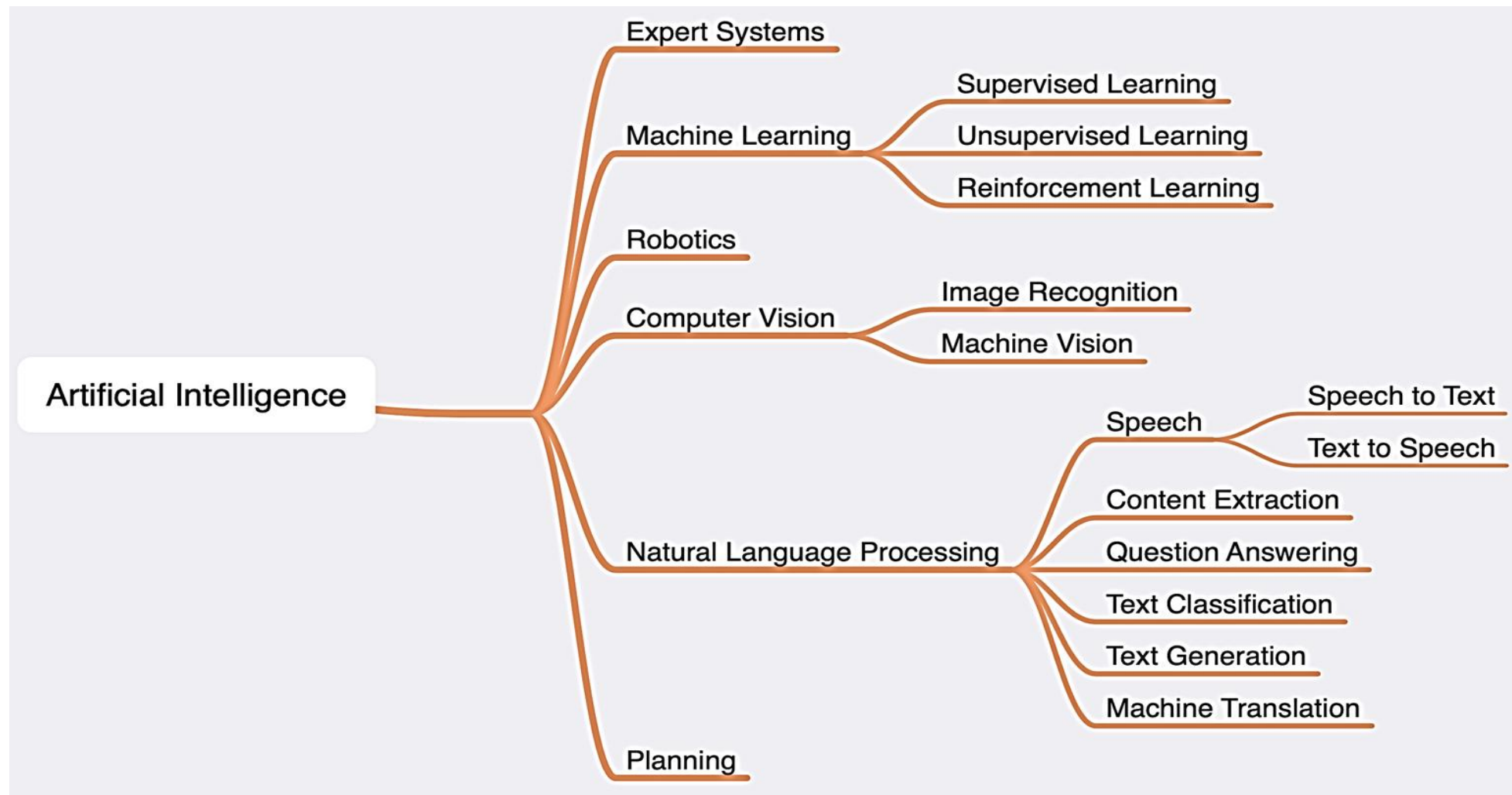


Russian Scientist Says...
'Immortality' Possible
...for Wealthy Elite by 2045



Russian media entrepreneur Dmitry Itskov is heading a project that will try and achieve human immortality within the next three decades (Image: 2045.com)

AI Branches



- Reasoning and decision making **(Main Focus of this course)**
 - Planning for Problem solving (search, decision theory)
 - Knowledge representation (Expert System, Fuzzy Logic)
 - Reasoning (logical, probabilistic, learning and action)

AI School of thoughts

- Symbolic AI (**GOF AI**) "Good Old-Fashioned Artificial Intelligence"
 - Started in 1956, failed in 80s.
 - Also known as symbolism.
 - Reasons logically from a set of facts and rules describing the domain.
 - Key advantage is that the reasoning process can be easily understood
 - A key disadvantage is that for learning, process has to be hand coded which is a hard problem.
 - Example: Expert Systems (based on rules), Turing Machine

AI School of thoughts

- Strong AI / Full AI (mimicking Human Brain)
 - an AI system can *think* and have a *mind, self improve*
 - There are no proper examples for Strong AI.
 - Initial Stage
- Weak AI / Narrow AI (No genuine intelligence)
 - an AI system can only *act* like it thinks and has a mind, limited ability to self improve
 - Examples: Google AI, Apple's Siri, Trading algorithms, An automatic car or remote control devices, computer games (Alpha Go), industrial automation.
 - Advanced Stage

AI School of thoughts

- **NEAT AI**

- **Neats** consider that solutions should be documentable and provably correct.
- Use logically supported, formal methods supported by statistics and built on transparent logic.
- Research Group supports:
 - Stanford, Carnegie Mellon University and the University of Edinburgh

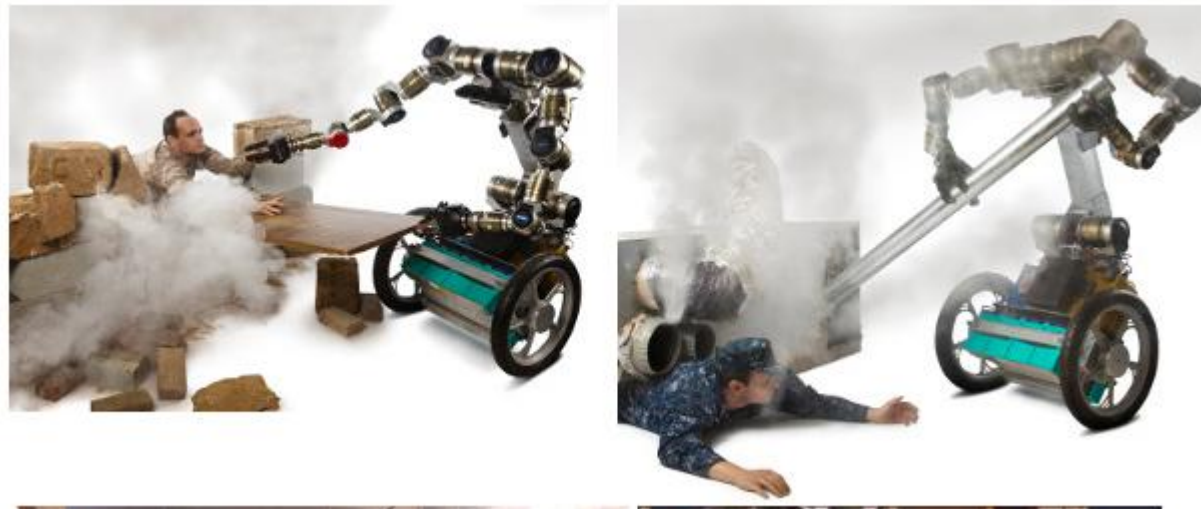
- **Scruffy AI**

- **Scruffies** believe that intelligence is too complicated to be solved with traditional software development style.
- Fuzzier and ambiguous methods that support results.
- Research Group supports:
 - MIT, and other universities Robotics research groups

- Neats vs. scruffies has also been described as “**logical versus analogical**” and “**symbolic versus connectionist**.”

AI Vs Cybernetics

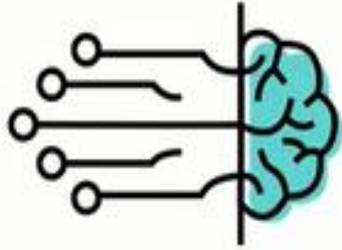
- **Cybernetics** is an old field, which is a subset of AI.
- AI nowadays is really broad, and spans Machine Learning, Statistics, Neurology, etc.
- Cybernetics is now a days included in **robotics** So in a sense cybernetics is a subset of AI.



AI School of thoughts

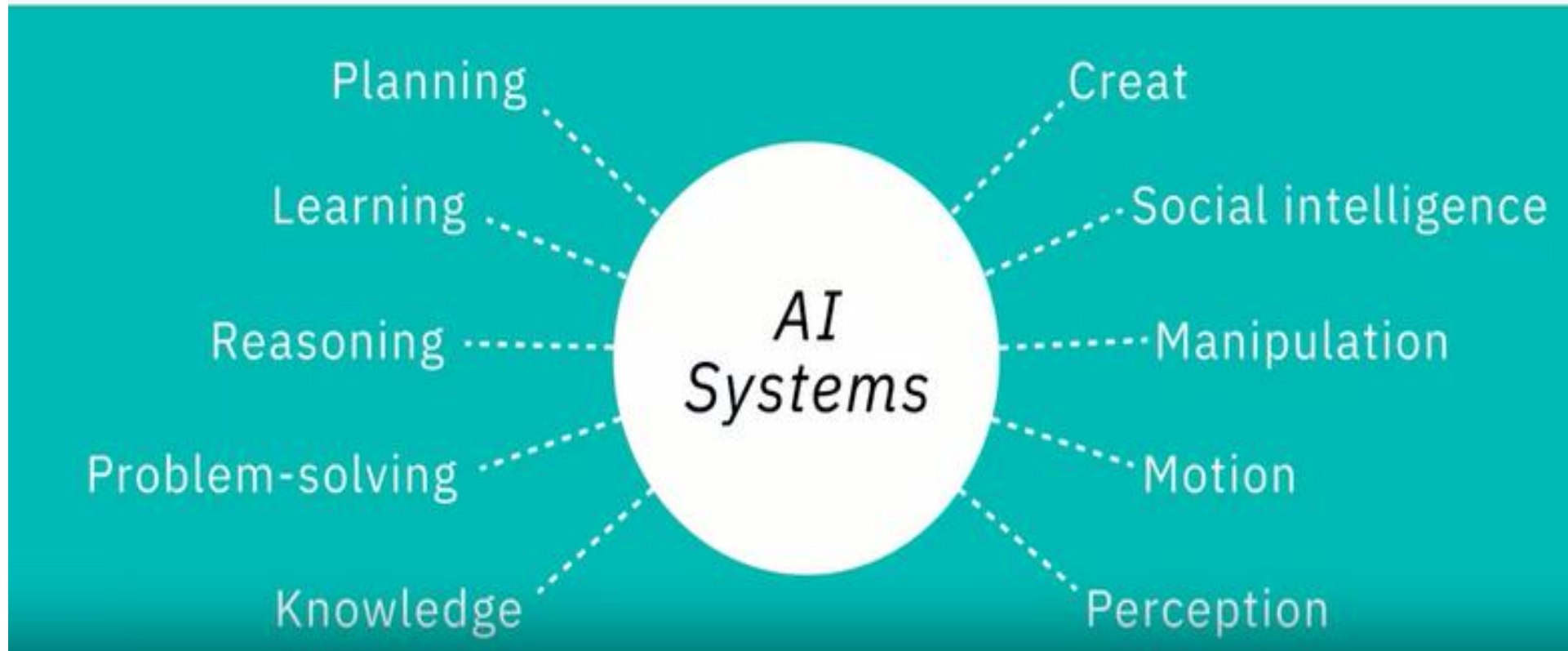
- **Artificial Life-> Soft, Hard, Wet**
- **“Soft” artificial life** creates computer simulations or other purely digital constructions that exhibit **life-like** behavior.
- **“Hard” artificial life** produces hardware implementations of **life-like** systems.
- **“Wet” artificial life** involves the creation of **life-like** systems in a laboratory using biochemical materials.

Summary



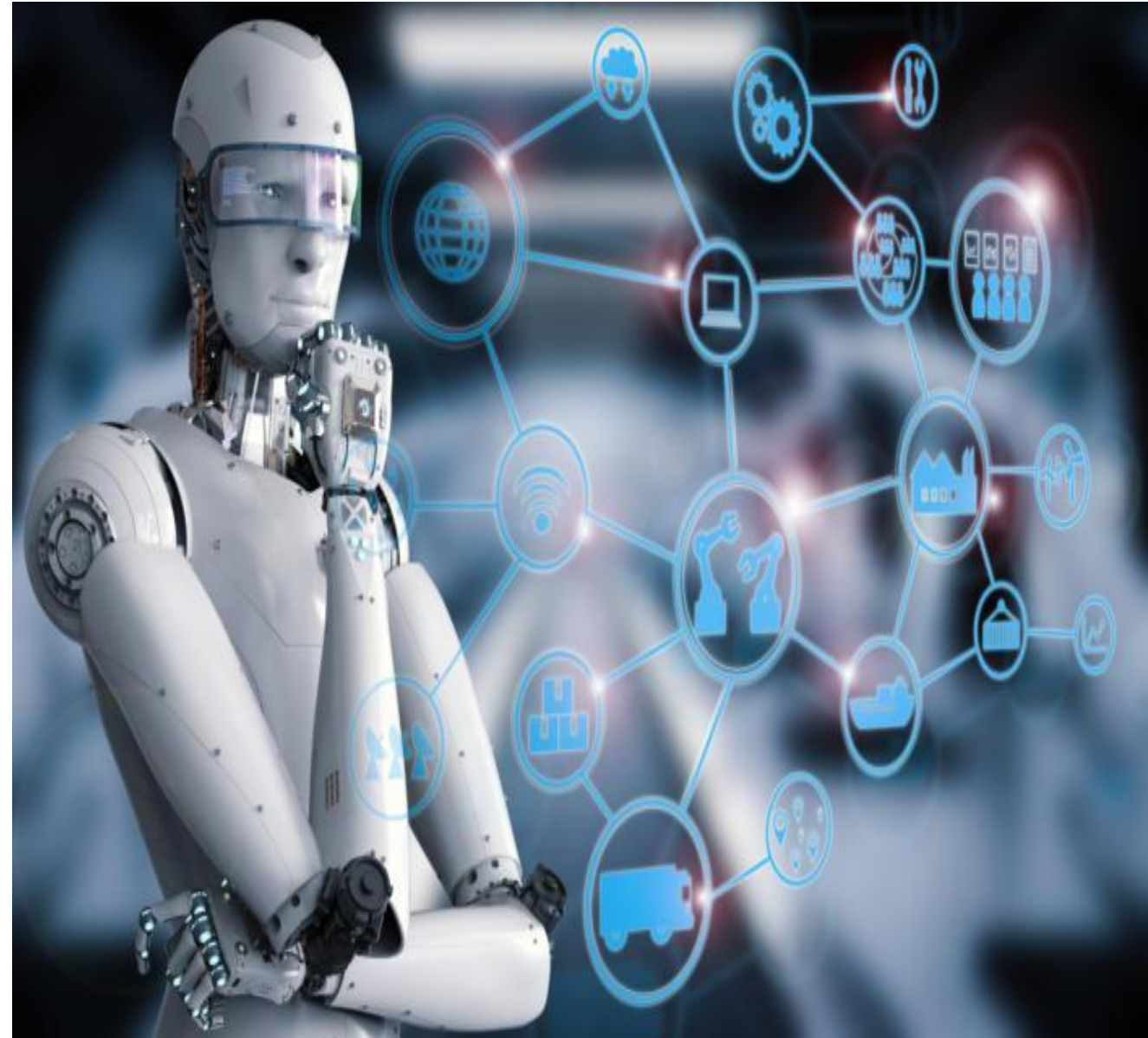
Artificial Intelligence

A branch of computer science dealing with the simulation of intelligent behavior



Summary of Today's Lecture

- Artificial Intelligence
- Introduction, History,
- Classification and Concepts
- AI Applications
- AI Research
- Future
- Assigned Reading
 - Chapter 1 in the text



Consider what might be involved in building an “intelligent” computer....

- What are the “components” that might be useful?
 - Fast hardware?
 - Foolproof software?
 - Chess-playing at grandmaster level?
 - Speech interaction?
 - speech synthesis
 - speech recognition
 - speech understanding
 - Image recognition and understanding ?
 - Learning?
 - Planning and decision-making?

Can we build hardware as complex as the brain?

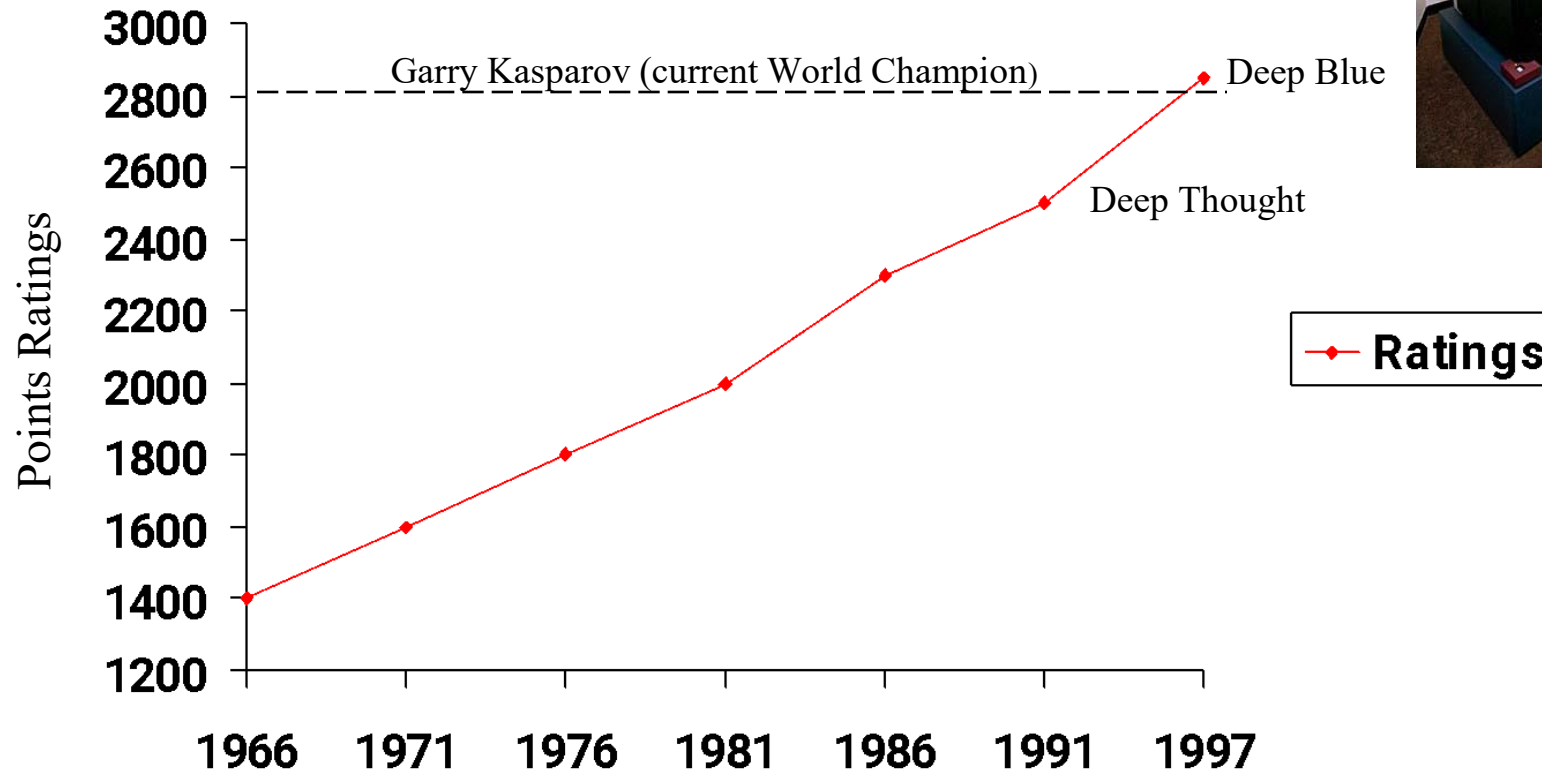
- How complicated is our brain?
 - a neuron, or nerve cell, is the basic information processing unit
 - estimated to be on the order of 10^{14} neurons in a human brain
 - cycle time: 10^{-3} seconds (1 millisecond)
- How complex can we make computers?
 - 10^6 or more transistors per CPU
 - supercomputer: hundreds of CPUs, 10^9 bits of RAM
 - cycle times: order of 10^{-8} seconds
- Conclusion
 - **YES**: in the near future we can have computers with as many basic processing elements as our brain, but with
 - far fewer interconnections (wires or synapses) than the brain
 - much faster updates than the brain
 - **but** building hardware is very different from making a computer behave like a brain!

Must an Intelligent System be Foolproof?

- A “foolproof” system is one that never makes an error:
 - Types of possible computer errors
 - hardware errors, e.g., memory errors
 - software errors, e.g., coding bugs
 - “human-like” errors
 - Clearly, hardware and software errors are possible in practice
 - what about “human-like” errors?
- An intelligent system can make errors and still be intelligent
 - humans are not right all of the time
 - we learn and adapt from making mistakes
- Conclusion:
 - **NO**: intelligent systems will not (and need not) be foolproof

Can Computers play Humans at Chess?

- Chess Playing is a classic AI problem
 - well-defined problem
 - very complex: difficult for humans to play well



- Conclusion: YES: today's computers can beat even the best human

Can Computers Talk?

- This is known as “speech synthesis”
 - translate text to phonetic form
 - e.g., “fictitious” -> fik-tish-es
 - use pronunciation rules to map phonemes to actual sound
 - e.g., “tish” -> sequence of basic audio sounds
- Difficulties
 - sounds made by this “lookup” approach sound unnatural
 - sounds are not independent of actions
 - a harder problem is emphasis, emotion, etc
- Conclusion: NO, for complete sentences, but YES for individual words

<https://www.youtube.com/watch?v=UWnuOqE4waw>

Can Computers Recognize Speech?

- Speech Recognition:
 - mapping sounds from a microphone into a list of words.
 - Hard problem: noise, more than one person talking, occlusion, speech variability,..
 - Even if we recognize each word, we may not understand its meaning.
- Recognizing single words from a small vocabulary
 - systems can do this with high accuracy (order of 99%)
 - e.g., directory inquiries
 - limited vocabulary (area codes, city names)
 - computer tries to recognize you first, if unsuccessful hands you over to a human operator
 - saves millions of dollars a year for the phone companies

Recognizing human speech (ctd.)

- Recognizing normal speech is much more difficult
 - speech is continuous
 - large vocabularies
 - background noise, other speakers, accents, colds, etc
 - on normal speech, modern systems are only about 60% accurate
- Conclusion: NO, normal speech is too complex to accurately recognize, but YES for restricted problems
 - (e.g., recent software for PC use by IBM, Dragon systems, etc)
 - **IBM Watson** is a question-answering computer system capable of answering questions posed in natural language, developed in IBM's **DeepQA** project by a research.

Can Computers Understand speech?

- Understanding is different to recognition:
- Conclusion: NO, much of what we say is beyond the capabilities of a computer to understand at present

Can Computers Learn and Adapt ?

- Learning and Adaptation

- consider a computer learning to drive on the freeway
- we could code lots of rules about what to do
- and/or we could have it learn from experience



Darpa's Grand Challenge. Stanford's "Stanley" drove 150 without supervision in the Mojave desert

- **machine learning** allows computers to learn to do things without explicit programming
- Conclusion: YES, computers can learn and adapt, when presented with information in the appropriate way

Can Computers “see”?

- Recognition v. Understanding (like Speech)
 - Recognition and Understanding of Objects in a scene
 - look around this room
 - you can effortlessly recognize objects
 - human brain can map 2d visual image to 3d “map”
- Why is visual recognition a hard problem?



- Conclusion: mostly NO: computers can only “see” certain types of objects under limited circumstances: but YES for certain constrained problems (e.g., face recognition)

Can Computers plan and make decisions?

- Intelligence
 - involves solving problems and making decisions and plans
 - e.g., plan a travel/ tour to Spain
- What makes planning hard?
 - the world is not predictable:
 - your flight is canceled or do you consider all flights? all dates?
 - **AI systems are only successful in constrained planning problems**
- Conclusion: NO, real-world planning and decision-making is still beyond the capabilities of modern computers
 - exception: very well-defined, constrained problems: mission planning for satellites.